



Katholieke
Universiteit
Leuven

Department of
Electrical Engineering

PROJECT 2: ANALYZING A CRYPTOSYSTEM BASED ON THE FINITE FIELD ISOMORPHISM PROBLEM

Computer Algebra for Cryptography (B-KUL-H0E74A)

Gust De Wit (r0948039)

Academic year 2025–2026

1 Finite Fields and Irreducible Polynomials

1.1 Finding an isomorphism

1.1.1 Alternate formulation

First, we will show that finding an isomorphism between $\mathbb{X}\langle x \rangle = \mathbb{F}_p[z]/f(z)$ and $\mathbb{Y}\langle y \rangle = \mathbb{F}_p[z]/g(z)$ is equivalent to finding a single root of $f(z)$ inside $\mathbb{Y}$.

Since f is monic of degree n over $\mathbb{F}_p$, we can write $f(z) = z^n + c_{n-1}z^{n-1} + \dots + c_0, \forall c_i \in \mathbb{F}_p$. Furthermore, we know that an isomorphism φ is a ring homomorphism and fixes $\mathbb{F}_p$. Hence, for any element $\alpha \in \mathbb{X}$:

$$\varphi(f(\alpha)) = \varphi(\alpha^n + c_{n-1}\alpha^{n-1} + \dots + c_0) = \varphi(\alpha)^n + c_{n-1}\varphi(\alpha)^{n-1} + \dots + c_0 = f(\varphi(\alpha)). \quad (1)$$

Applying this to $\alpha = x$ and using $f(x) = 0$ by definition of $\mathbb{X}$, we get $0 = \varphi(0) = \varphi(f(x)) = f(\varphi(x))$ in $\mathbb{Y}$. **Hence $\varphi(x)$ is a root of f in $\mathbb{Y}$.**

“Conversely, let $y_0 \in \mathbb{Y}$ be any root of f . Every element of $\mathbb{X}$ can be written uniquely as $a(x)$ for a polynomial a of degree $< n$ over $\mathbb{F}_p$; define $\varphi(a(x)) = a(y_0)$ for every such element. This map is well defined (if $f \mid a - b$ then $a(y_0) - b(y_0) = 0$ since $f(y_0) = 0$), it is a ring homomorphism, and it is injective since $\mathbb{X}$ is a field; as $|\mathbb{X}| = |\mathbb{Y}| = p^n$, it is a bijection, hence an isomorphism. Therefore determining an isomorphism is exactly determining one root y_0 of f in $\mathbb{Y}$.”

1.1.2 Existence and number of roots of f in $\mathbb{Y}$

By construction $\mathbb{X} = \mathbb{F}_p[z]/(f)$ is $\mathbb{F}_{p^n}$, a degree- n Galois extension of $\mathbb{F}_p$ with cyclic group generated by the Frobenius $\sigma(u) = u^p$. The generator x satisfies $f(x) = 0$ with f monic irreducible of degree n , so f is its minimal polynomial; since $\deg f = n$, x lies in no proper subfield, hence $f = \prod_{i=0}^{n-1} (z - \sigma^i(x)) = \prod_{i=0}^{n-1} (z - x^{p^i})$. The n conjugates are distinct (otherwise x would lie in a proper subfield), so f has exactly n distinct roots, all in $\mathbb{F}_{p^n}$. As $\mathbb{Y} \cong \mathbb{F}_{p^n}$, f likewise splits completely over $\mathbb{Y}$ into n distinct linear factors, so the algorithm only ever extracts linear factors.

1.1.3 Algorithm

Since all of the factors are linear, we can skip the square-free factorisation and distinct-degree factorisation steps and move straight to the equal-degree factorisation step (Cantor-Zassenhaus algorithm), with the factor degree known to be 1.

Algorithm 1 FINDROOT(f) — one root of f in $\mathbb{Y}$

Require: monic $f \in \mathbb{Y}[w]$ splitting into distinct linear factors over $\mathbb{Y}$

Ensure: a root $y_0 \in \mathbb{Y}$ of f

```

1: while  $\deg f > 1$  do
2:   repeat
3:      $h \leftarrow \text{CANTORZASSENHAUS}(f)$ 
4:   until  $h \neq \text{FAILURE}$ 
5:    $f \leftarrow h$  ▷ descend into one proper factor
6: end while
7: make  $f$  monic; write  $f = w - y_0$ 
8: return  $y_0$ 

```

This algorithm repeatedly calls the Cantor-Zassenhaus algorithm as described on slide 27 of the lecture notes, without the extra step of calculating the gcd because in our case, we know all roots like in $\mathbb{F}_q$. Each subsequent call to Cantor-Zassenhaus returns a factor of the submitted $f(x)$ or FAILURE with chance $\leq 1/2$. This factor is not guaranteed to be linear yet, so we repeat this procedure until it is. At the end, we make the f_i monic and calculate y_0 .

1.1.4 Complexity

Let $M(n)$ be the cost in $\mathbb{F}_p$ -operations of multiplying two degree- $< n$ polynomials over $\mathbb{F}_p$. One multiplication in $\mathbb{Y}$ costs $O(M(n))$, so one in $\mathbb{Y}[w]/(f)$ costs $O(M(n)^2)$. Root finding over a field of size Q costs $O(M(n) \log n \log(nQ))$ operations in that field; here the field is $\mathbb{Y}$ with $Q = p^n$, so $\log(nQ) = \Theta(n \log p)$. Converting each $\mathbb{Y}$ -operation to $\mathbb{F}_p$ -operations (a factor $O(M(n))$) yields

$$O(n \log p \cdot \log n \cdot M(n)^2)$$

$\mathbb{F}_p$ -operations, i.e. polynomial in n and $\log p$.

1.1.5 Finding other isomorphisms

The cost of finding other isomorphisms is almost free once we have one root y_0 , as the remaining $n - 1$ roots of f in $\mathbb{Y}$ are its Frobenius conjugates $y_0, y_0^p, \dots, y_0^{p^{n-1}}$. This requires only exponentiation.

1.2 Implementation of FieldIsomorphism(f, g)

An implementation of FIELDISOMORPHISM can be found in `field_iso.m`. The results of running the program with $p = 2^{16} + 7$ and $n = 2^k$ for $k = 2, \dots, 6$ are reported in Table 1.

n	4	8	16	32	64
time (s)	0.000	0.020	0.090	0.860	10.110

Table 1: Running time of FIELDISOMORPHISM for $p = 2^{16} + 7$ and $n = 2^k$, $k = 2, \dots, 6$.

1.3 Implementation of InverseFieldIsomorphism(f, g, s)

An implementation of INVERSEFIELDISOMORPHISM can be found in `field_iso.m`. This algorithm relies on the fact that φ is not only a field isomorphism, but also an $\mathbb{F}_p$ -linear map between the two $\mathbb{F}_p$ -vector spaces $\mathbb{X}$ and $\mathbb{Y}$ with basis $1, x, x^2, \dots, x^{n-1}$ and $1, y, y^2, \dots, y^{n-1}$ respectively. Since this is a linear map, defined by what it does to a basis, it can be described as a matrix.

We know $\varphi = s(y) = y_0$, and since φ is a ring homomorphism, $\varphi(x^i) = \varphi(x)^i = y_0^i$. So the image of the i -th basis vector of $\mathbb{X}$ is $y_0^i \in \mathbb{Y}$. We can write each y_0^i as its coordinate vector in $\mathbb{Y}$'s basis, resulting in an $n \times n$ matrix M over $\mathbb{F}_p$.

Since φ is a bijection, M is invertible, and M^{-1} is the matrix of the inverse map $\psi = \varphi^{-1}$. Applying M^{-1} to the generator y yields a polynomial t in $\mathbb{X}$'s basis, with $\psi(y) = t(x)$.

2 A Cryptosystem Based on the FFIP

The system as described in the assignment was implemented in `ffip.m`.

3 Attacking the FFIP Cryptosystem (the stupid way)

3.1 Known plaintext-noise-ciphertext-attack

3.1.1 The attack

Recall from Section 1.3 that ϕ is an $\mathbb{F}_p$ -linear map $\mathbb{X} \rightarrow \mathbb{Y}$, represented by an invertible $n \times n$ matrix M over $\mathbb{F}_p$ with respect to the bases $\{1, x, \dots, x^{n-1}\}$ and $\{1, y, \dots, y^{n-1}\}$. Our attack will recover s from M directly from this linear structure. For every sample, $c_i = M * w_i := M * (m_i + 2r_i) = \sum_{j=0}^{n-1} w_{i,j} x^j$, with $w_{i,j}$ in $\mathbb{F}_p$ and c_i coordinate column vectors of length n . Stack the n samples as columns: $C = M * W$, with $W = [w_1 | w_2 | \dots | w_n]$ and $C = [c_1 | c_2 | \dots | c_n]$. If W is invertible over $\mathbb{F}_p$, then $M = C * W^{-1}$.

From M , we recover $\phi(x)$ as the image of the basis vector $x = [0, 1, 0, \dots, 0]^T$, whose image is exactly the coefficient vector $s(z)$.

3.1.2 Probability of failure

The probability that this attack fails is the probability that W is singular over $\mathbb{F}_p$. This is dependent on the distribution of the random noise added. The values w_i can be very limited, with entries $m_{i,j} \in \{0, 1\}$ and $r_{i,j} \in \{-1, 0, 1\}$. This results in only 6 possible values for w_i : $\{-2, -1, 0, 1, 2, 3\}$, all with the chance of occurring and independent of each other, making all entries of W i.i.d. uniform on a 6-element subset of $\mathbb{F}_p$. The chance of this resulting in a singular matrix is the result of Kahn and Komlós (2001). This paper states

$$\Pr(M \text{ is non-singular}) \rightarrow \prod_{i \geq 1} (1 - p^{-i}) \quad \text{as } n \rightarrow \infty$$

as an asymptotic limit to the non-singular case, valid for our fixed, non-degenerate distribution μ — the uniform distribution on the support $\{-2, -1, 0, 1, 2, 3\} \subseteq \mathbb{F}_p$. We’re looking for the probability that it *is* singular, so the probability that our attack fails is $1 - \Pr(M \text{ is non-singular})$.

3.1.3 The underdetermined case

With $t < n$ triples, the linear system for M is underdetermined. We can restore uniqueness by exploiting that ϕ is also multiplicative, hence, *all* of M can be derived from a single, leaving only n unknowns. Each triple gives n scalar equations $c_i = \sum_j w_{i,j} s(y)^j$, but these are now nonlinear in the n unknowns. This system can be solved by making use of a Gröbner basis.

3.1.4 Recovering $f(z)$ from ϕ

“ Once the isomorphism ϕ is known as the polynomial $s(z)$ (so $\phi(x) = s(y) \in \mathbb{Y}$), recovering $f(z)$ reuses the argument of Section 1.1.1. There we showed $\phi(f(x)) = f(\phi(x))$ and hence that $\phi(x)$ is a root of f in $\mathbb{Y}$; equivalently $f(s(y)) = 0$. Since ϕ is a bijection (Section 1.3), $s(y)$ generates $\mathbb{Y}$ over $\mathbb{F}_p$ and therefore lies in no proper subfield, so by the same minimal-polynomial reasoning as in Section 1.1.2 its minimal polynomial over $\mathbb{F}_p$ has degree exactly n and equals $f(z)$.

We compute it by linear algebra over $\mathbb{F}_p$. Reducing modulo $g(z)$, form the coordinate vectors of $1, s(y), s(y)^2, \dots, s(y)^n$ in the basis $\{1, y, \dots, y^{n-1}\}$ of $\mathbb{Y}$. These $n + 1$ vectors lie in an n -dimensional space, so they admit a linear dependency $\sum_{k=0}^n a_k s(y)^k = 0$; it is unique up to scaling because $s(y)$ has a degree- n minimal polynomial. Normalising so that $a_n = 1$ yields $f(z) = \sum_{k=0}^n a_k z^k$. ”

3.2 Implementation of KnownPlaintextNoiseAttack(g, cs, ms, rs)

An implementation of the attack can be found as `KNOWNPLAINTEXTNOISEATTACK(G, CS, MS, RS)`. The results of running the tests can be found in Table 2. The estimated failure rate according to Kahn and Komlós (2001) is also displayed, this approximation would get better for larger n .

3.3 Gröbner basis attack from a single ciphertext

3.3.1 Setup

We are now given only one ciphertext-plaintext pair (c, m) , but not the noise r . Encryption is $c = \phi(m + 2r)$, so the linear attack of the previous section no longer applies. Instead we follow the non-lattice algebraic attack of Doröz et al. (2017, §2.4.2), which avoids working with ϕ directly by using that it is a field isomorphism.

Applying the inverse isomorphism $\psi = \phi^{-1}$ to the ciphertext gives, inside $\mathbb{X} = \mathbb{F}_p[z]/(f(z))$, $\psi(c) = m + 2r$, i.e. $c(z) \equiv m(z) + 2r(z) \pmod{f(z)}$.

Since $\deg(c - m - 2r) < n = \deg f$, this is the polynomial identity $c(z) = Q(z)f(z) + m(z) + 2r(z)$ in $\mathbb{F}_p[z]$ for an auxiliary quotient Q . The unknowns are the coefficients of r (the noise, in $\{-1, 0, 1\}$), of f (which has

k	p	avg. time (s)	succeeded	fail-rate (emp.)	fail-rate (pred.)
1	3	0.012	585/1000	0.4150	0.4399
2	5	0.012	776/1000	0.2240	0.2397
3	11	0.012	898/1000	0.1020	0.0992
4	17	0.012	949/1000	0.0510	0.0623
5	37	0.012	974/1000	0.0260	0.0278
6	67	0.012	976/1000	0.0240	0.0151
7	131	0.012	990/1000	0.0100	0.0077
8	257	0.013	998/1000	0.0020	0.0039

Table 2: Known-plaintext-with-noise attack (Task 3.b) for $n = 32$, $p = \text{NextPrime}(2^k)$, 1000 trials per prime. “Succeeded” counts runs recovering the exact secret $s(z)$; the empirical failure rate is compared against the Kahn-Komlós prediction $1 - \prod_{i \geq 1} (1 - p^{-i})$.

the special shape $f = z^n + h(z)$ with $\deg h \leq \lfloor n/2 \rfloor$ and coefficients of h in $\{-1, 0, 1\}$, and of Q . Matching coefficients on both sides yields a system of non-linear polynomial equations.

This leads to an underdetermined system of equations, and after eliminating the isomorphism unknowns by a Gröbner basis, we are left with the problem of finding a small point on an n -dimensional variety. This is where the paper of Doröz et al. (2017) stops.

3.3.2 From an n -dimensional to a zero-dimensional variety

We resolve this by encoding smallness as equations rather than as a search condition, taking inspiration from Arora and Ge (2011). A value lies in $\{-1, 0, 1\}$ *if and only if* it is a root of $t^3 - t = t(t-1)(t+1)$. We therefore add, for every unknown coefficient r_j of the noise and every unknown coefficient h_k of f , the cubic equation

$$r_j^3 - r_j = 0, \quad h_k^3 - h_k = 0.$$

This adds roughly $3n/2$ cubic equations, each confining its unknown to a 3-element set. The common solution set of the divisibility equations together with these cubic equations is therefore *finite*, i.e. a zero-dimensional variety, instead of the n -dimensional variety of Doröz et al. (2017). Using Buchberger’s algorithm, a Gröbner basis then returns the finitely many candidate pairs (f, r) , from which we keep the one whose f has the correct monic shape.

3.3.3 Complexity

Once the system is zero-dimensional, the cost of the Gröbner basis computation is governed by the degree of regularity d_{reg} . With $N = \Theta(n)$ unknowns and $m = \Theta(n)$ equations, the standard estimate is

$$\mathcal{O}\left(\left(m \binom{N + d_{\text{reg}} - 1}{d_{\text{reg}}}\right)^\omega\right),$$

with ω the linear-algebra exponent ($\omega \approx 3$ in practice). For a semi-regular system d_{reg} is the index of the first non-positive coefficient of the Hilbert series

$$H(z) = \frac{\prod_i (1 - z^{d_i})}{(1 - z)^N},$$

where the dominating high-degree generators are the cubic smallness equations ($d_i = 3$). However, because the smallness equations make the system strongly overdetermined, d_{reg} stays small, which is why the attack is feasible only for moderate n .

3.4 Implementation of GrobnerBasisAttack(g, c, m, α, β)

An implementation of the attack can be found in GROBNERBASISATTACK(G, C, M, α, β). The attack was tested against $p = 2^{16} + 7$, $\alpha = -1$, $\beta = 1$ and $n = 2, 3, 4$ and the results can be found in Table 3.

n	avg. time (s)	avg. #valid keys	valid	equals secret
2	0.001	2.7	10/10	5/10
3	0.005	1.9	10/10	7/10
4	0.356	1.4	10/10	7/10

Table 3: Single-ciphertext Gröbner-basis attack (Task 3.d) for $p = 2^{16} + 7$, $\alpha = -1$, $\beta = 1$, 10 trials per n . “valid” counts trials where the returned (s, f) is a working key (f irreducible of degree n , s a root of f); “equals secret” counts trials where it is the original key. “avg. #valid keys” is the mean number of distinct valid keys consistent with the single ciphertext.

3.4.1 Comparison against exhaustive search

We’re looking for the secret pair (f, s) , however, s is recoverable from f , analogous to Section 1.1. So really we’re only looking for $f = z^n + h(z)$ with degree $h \leq \lfloor n/2 \rfloor$ and each coefficient of h in $\{-1, 0, 1\}$. So the searchspace for f is $3^{\lfloor n/2 \rfloor + 1}$, and for each candidate we must: (i) test irreducibility, (ii) construct the isomorphism s ($\mathcal{O}(n \log p \cdot \log n \cdot M(n)^2)$), (iii) check it’s consistent with the known pair (c, m) by verifying the smallness of r .

Compared to Gröbner, at the tested sizes ($n = 2, 3, 4$) neither is significantly faster. However, their asymptotic behaviour does differ, and at large n , if $d_{\deg}$ stays small, Gröbner won’t scale exponentially while exhaustive search will.

3.4.2 Correctness

Also visible in Table 3 is that we do not always get back the correct secret key. This is because, by chance, multiple possible solutions fit the constraints layed out by the equations. In practice, this could be minimized by adding a second ciphertext-plaintext pair encrypted with the same key and only accepting a key which worked on both. This practically rules out the existence of other correct solutions.

4 Breaking Search FFIP (the clever way)

4.1 Representation-independence of the trace

For $a \in \mathbb{F}_{p^n}$ the trace is $\text{Tr}(a) = \sum_{k=0}^{n-1} a^{p^k} = \sum_{k=0}^{n-1} \sigma^k(a)$, the sum over the Frobenius orbit, with $\sigma(u) = u^p$ the Frobenius generator of the Galois group already used in Section 1.1.2. We show that for any field isomorphism $\varphi : \mathbb{X} \rightarrow \mathbb{Y}$ and any $a \in \mathbb{X}$,

$$\text{Tr}_{\mathbb{Y}}(\varphi(a)) = \text{Tr}_{\mathbb{X}}(a). \quad (2)$$

Write $\sigma_{\mathbb{X}}(u) = u^p$ and $\sigma_{\mathbb{Y}}(v) = v^p$ for the Frobenius maps on $\mathbb{X}$ and $\mathbb{Y}$. Both are the canonical generator of the respective Galois group: the map $u \mapsto u^p$ is a ring homomorphism in characteristic p (it respects products trivially, and $(u + v)^p = u^p + v^p$ since $\binom{p}{k} \equiv 0 \pmod{p}$ for $0 < k < p$), it is injective hence bijective on the finite field, and by Fermat its fixed field is exactly $\mathbb{F}_p$.

The single ingredient is that φ commutes with Frobenius. As an isomorphism φ is a ring homomorphism, so in characteristic p

$$\varphi(u^p) = \varphi(\underbrace{u \cdots u}_p) = \varphi(u)^p, \quad (3)$$

that is $\varphi \circ \sigma_{\mathbb{X}} = \sigma_{\mathbb{Y}} \circ \varphi$, and iterating k times gives $\varphi(a^{p^k}) = \varphi(a)^{p^k}$ for every k . Using additivity of φ ,

$$\varphi(\text{Tr}_{\mathbb{X}}(a)) = \varphi\left(\sum_{k=0}^{n-1} a^{p^k}\right) = \sum_{k=0}^{n-1} \varphi(a^{p^k}) = \sum_{k=0}^{n-1} \varphi(a)^{p^k} = \text{Tr}_{\mathbb{Y}}(\varphi(a)). \quad (4)$$

Finally $\text{Tr}_{\mathbb{X}}(a) \in \mathbb{F}_p$, since raising it to the p -th power only cyclically permutes the n summands, so it is fixed by $\sigma_{\mathbb{X}}$ and the fixed field is $\mathbb{F}_p$. Every isomorphism fixes the prime field pointwise (a ring homomorphism

sends $1 \mapsto 1$, hence fixes $0, 1, 1+1, \dots$, so $\varphi(\text{Tr}_{\mathbb{X}}(a)) = \text{Tr}_{\mathbb{X}}(a)$. Combining with the previous equation, $\text{Tr}_{\mathbb{X}}(a) = \varphi(\text{Tr}_{\mathbb{X}}(a)) = \text{Tr}_{\mathbb{Y}}(\varphi(a))$. Intuitively, the trace depends only on the abstract element and its Frobenius conjugates; an isomorphism carries conjugates to conjugates while fixing $\mathbb{F}_p$, so the sum is unchanged regardless of the representation.

References

- Arora, Sanjeev, and Rong Ge. 2011. *New Algorithms for Learning in Presence of Errors*, 403–415. January. https://doi.org/10.1007/978-3-642-22006-7_{_}34. https://doi.org/10.1007/978-3-642-22006-7_34.
- Doröz, Yarkin, Jeffrey Hoffstein, Jill Pipher, Joseph H. Silverman, Berk Sunar, William Whyte, and Zhenfei Zhang. 2017. *Fully Homomorphic Encryption from the Finite Field Isomorphism Problem*. Cryptology ePrint Archive, Paper 2017/548. <https://eprint.iacr.org/2017/548>.
- Kahn, Jeff, and János Komlós. 2001. “Singularity Probabilities for Random Matrices Over Finite Fields.” *Combinatorics Probability Computing* 10, no. 2 (March): 137–157. <https://doi.org/10.1017/s096354830100462x>. <https://doi.org/10.1017/s096354830100462x>.